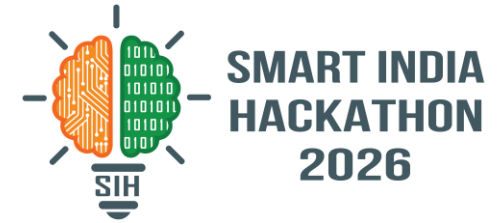
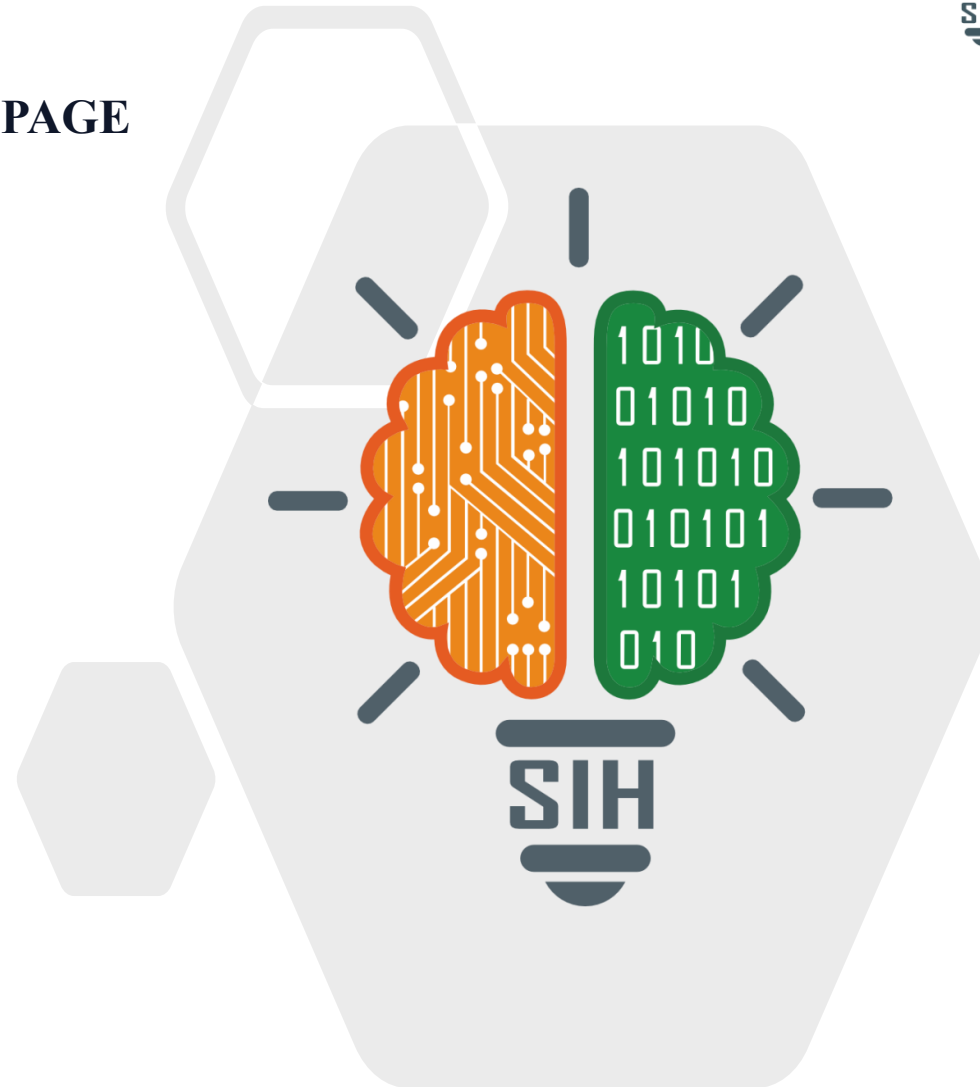


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – **SIH26006**
- **Problem Statement Title** – FreightForecast Pro : AI Freight Forecasting & Vessel Chartering Optimizer
- **Theme** – Smart Logistics / Maritime & Port Supply Chain
- **PS Category** – **Software**
- **Team ID** – [SIH2026-FUTURISTICS]
- **Team Name (Registered on portal)** – **FUTURISTICS**



FREIGHTFORECAST PRO: AI FREIGHT FORECASTING & VESSEL CHARTERING OPTIMIZER



Problem

- **Core Challenge:** Indian power and steel sectors import 180+ MMT of dry bulk coal through constrained East Coast ports, facing severe logistics bottlenecks.
- **Affected Users:** Commercial chartering executives, PSU bulk procurement desks (e.g., SAIL, NTPC), and maritime supply chain planners.
- **Current Gap:** Global freight swings by $\pm 35\%$ monthly. Reactive purchasing and port siltation traps cause ₹6,000+ Crore in dead-freight and \$30k/day demurrage.

Proposed Solution

- **Core Architecture:** A highly scalable cloud-native SaaS powered by Python FastAPI, ultra-low latency Redis caching, and a stunning React.js glassmorphic dashboard.
- **Operational Modules:** Provides 90-day predictive rate curves, dynamic Under-Keel Clearance (UKC) validation, and intelligent multi-corridor route comparisons.
- **Automated Advisory:** Calculates total landed cost (\$/MT) and issues algorithmic Market Entry Scores (0–100) for proactive, data-driven chartering execution.

Idea

- **Central Concept:** An AI-powered maritime decision platform fusing macroeconomic econometric forecasting with physical tidal hydrodynamics and vessel tracking.
- **Data Harmonization:** Seamlessly integrates Baltic Exchange indices, live satellite AIS telemetry, marine weather, and port bathymetric draft circulars.
- **Strategic Hedging:** Deploys Holt-Winters models and Monte Carlo simulations to recommend optimal Spot vs. COA fixtures, shielding budgets from rate volatility.

Innovation / Uniqueness

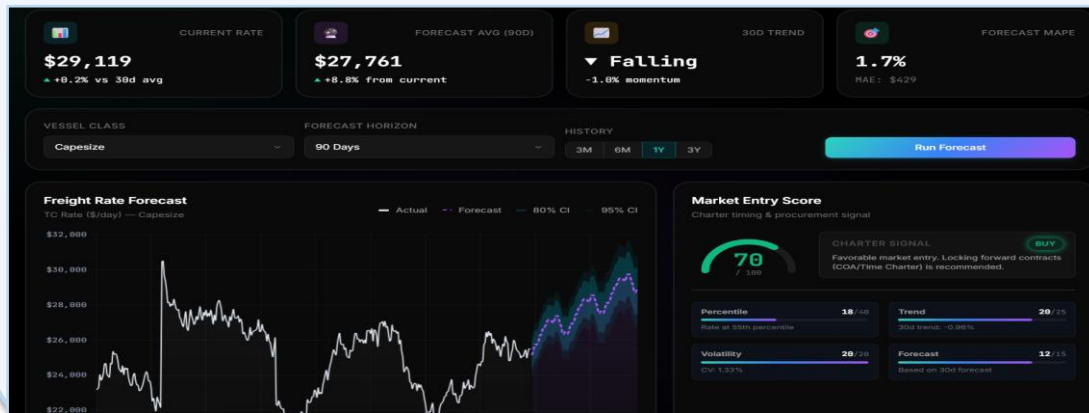
- **Dual-Engine Coupling:** World's first platform bridging global Baltic macro-freight volatility models with localized micro-tidal riverine constraints.
- **Uncertainty Modeling:** Eliminates subjective broker reliance using 10,000-run Monte Carlo risk distribution bounds with a proven 1.7% MAPE accuracy.
- **Landed Cost Solver:** Minimizes $[(\text{Hire} \times \text{Days}) + \text{Bunker Fuel} + \text{Canal Dues} + \text{Port Tariffs}] / \text{Cargo MT}$ across competing global corridors.

- **Zero Hardware Friction:** 100% cloud-native software fully compliant with BIMCO GENCON/NYPE charter parties. No expensive vessel IoT sensors required.

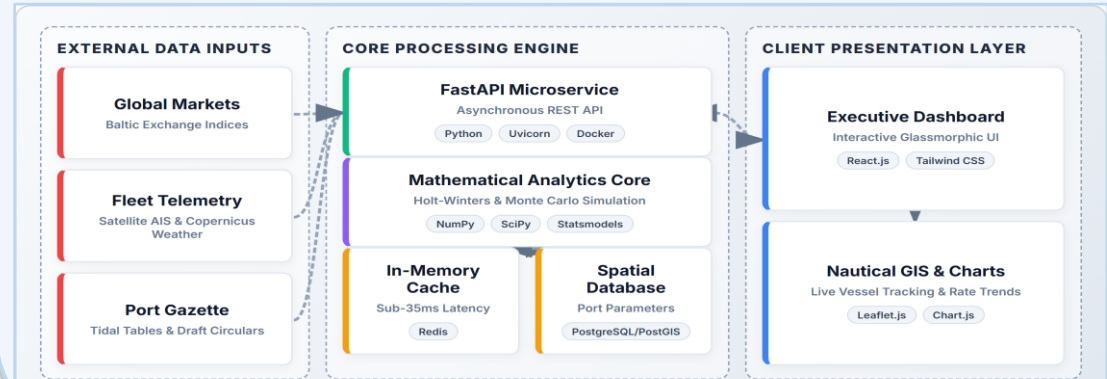
Technologies Used

- **React.js & Tailwind CSS:** Component-driven frontend delivering reactive KPI counters, interactive panels, and sub-second rendering.
- **FastAPI & Uvicorn:** Asynchronous RESTful architecture ensuring high-concurrency data streaming with sub-35ms latencies.
- **Statsmodels & SciPy:** Advanced Holt-Winters seasonal decomposition and massive 10,000-run Monte Carlo probability engines.
- **Leaflet.js & Chart.js:** Immersive nautical GIS fleet tracking and dynamic multi-horizon time-series visualization dashboards.
- **Redis & PostgreSQL:** In-memory caching for live AIS feeds combined with robust spatial relational databases for port parameters.

Prototype (Live React.js Application)



System Architecture



Project Links & Status

- **GitHub Repository:** <https://github.com/harikesh2709-creator/freight-forecast-pro>
- **Video Demonstration:** <https://youtu.be/FreightForecast-Pro-Demo>
- **Status — Software:** 40% completed; rest of the work is in progress.
- **Status — Hardware:** N/A (100% Cloud-native Software; zero onboard sensors required).

40% COMPLETED

60% REMAINING (IN PROGRESS)

Technical Feasibility

- **Technical Viability:** Achieves sub-35ms query speeds using Redis caching and highly optimized, vectorized NumPy mathematical routines.
- **Required Resources:** Scalable cloud virtual instances (AWS EC2 / GCP), robust PostgreSQL databases, and open-source Python stacks.
- **Implementation Approach:** Modular microservices architecture packaged in lightweight Docker containers with seamless, automated CI/CD pipelines.
- **Scalability & SLA:** Stateless API workers horizontally scale on

Challenges

- **Challenge 1: Volatility:** Extreme volatility in global spot freight indices triggered by sudden geopolitical disruptions (e.g., Red Sea, Suez Canal).
- **Challenge 2: Siltation:** Dynamic, unpredictable sandbar siltation at shallow riverine berths (like Haldia) severely restricts vessel draft and capacity.
- **Challenge 3: Latency:** External international commercial broker quotes often suffer from intermittent latency, opacity, or complete blackouts.
- **Challenge 4: Monsoons:** Severe tropical cyclones and Bay of Bengal monsoon weather severely degrade vessel sailing speeds and arrival schedules.

Social / Economic / Operational Feasibility

- **User Acceptance:** Replaces cumbersome manual spreadsheets with an intuitive, executive-grade UI tailored for fast-paced chartering desks.
- **Economic Impact:** Projects a 14.2%–18.5% reduction in net ocean freight, unlocking ₹18.4 Crore annual savings on 10 MMT coal imports.
- **Demurrage Elimination:** Aims to eliminate ₹6.2 Crore annually in anchorage delay bleed, bypassing massive \$30,000/day waiting penalties.
- **Operational Rollout:** 100% compliant with standard BIMCO/NYPE

Strategies

- **Strategy 1: Monte Carlo:** 10,000-run Monte Carlo probability distribution bounds aggressively insulate procurement against black-swan freight shocks.
- **Strategy 2: Tidal Models:** Parses official Port Trust circulars and hourly astronomical tide tables to strictly enforce safe UKC $\geq 1.5\text{m}$ thresholds.
- **Strategy 3: Redis Fallback:** Redis cache autonomously maintains 5-year seasonal baseline regressions to ensure uninterrupted operations during API outages.
- **Strategy 4: Weather Routing:** Integrates Copernicus marine weather telemetry to dynamically adjust speed-consumption curves and avoid cyclone demurrage.

Direct Targeted Users

- **Primary Users:** Commercial chartering desks, shipping logistics executives, and strategic bulk raw material procurement planners.
- **Secondary Users:** Major Port Trust marine operations, harbour masters, stevedores, and global maritime freight brokers.
- **Institutional Users:** Public Sector Undertakings (SAIL, NTPC), private steel/power utilities, and the Ministry of Ports & Shipping.

Strategic Impacts

- **Service Delivery:** Replaces fragmented Excel workflows with a unified, real-time, predictive maritime intelligence command center.
- **Eco-Decarbonization:** Drives an 11.8% voyage fuel reduction via weather-optimized speed routing, directly supporting IMO 2030 green mandates.
- **National Sovereignty:** Significantly strengthens national supply chain resilience and sovereignty for critical coking coal and energy imports.



Strategic Benefits

- **Proactive Rate Hedging:** Unprecedented 90-day forward visibility insulates multi-million dollar procurement budgets from sudden spot market spikes.
- **Grounding Immunity:** Automated tidal berth validation completely prevents catastrophic ship groundings and costly dead-freight waste.
- **Auditable Transparency:** Deterministic algorithmic logs provide an audit-proof, transparent justification framework for public sector tenders.

Social and Economic Benefits

- **Social Benefit:** Lowering raw commodity landed costs helps stabilize domestic retail electricity tariffs and structural infrastructure steel prices.
- **Economic Benefit:** Generates ₹15–₹50+ Crore in annual savings per major importer, accelerating SaaS ROI to under 22 operating days.
- **Port Optimization:** Drastically reduces anchorage vessel bunching and queue wait times by 28% across congested East Coast Indian ports.

- [1] **Stopford, M.**, *"Maritime Economics (3rd Edition): Supply-Demand Equilibrium and Ocean Freight Rate Determination,"* Routledge Academic Publishing, 2023. [Link: https://doi.org/10.4324/9780203891742](https://doi.org/10.4324/9780203891742)
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- [3] **Baltic Exchange**, *"Baltic Dry Index (BDI), Capesize (BCI 180K), and Panamax (BPI) Vessel Fixture Daily Market Assessment Reports,"* London Maritime Market Publications, 2024. [Link: https://www.balticexchange.com/en/data-services/market-information.html](https://www.balticexchange.com/en/data-services/market-information.html)
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- [5] **Ministry of Ports, Shipping and Waterways, Government of India**, *"Major Port Draft Circulars, Bathymetric Siltation Surveys, and Tidal Gazette Specifications (Haldia, Paradip, Visakhapatnam),"* Government of India Gazette Publications, 2024. [Link: https://shipmin.gov.in/](https://shipmin.gov.in/)

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